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RESEARCH ARTICLE

A face mask detection system: An approach to fight with COVID-19 scenario

Ruchi Jayaswal[1,2] | **Manish Dixit**[1]

[1]Dept. of CSE/IT, MITS, Gwalior, India
[2]Dept. of CSE/IT/AIML,Symbiosis Institute of Technology,Pune,India

Correspondence
Ruchi Jayaswal, Dept. of CSE/IT, MITS, Gwalior (MP.), India.
Email: ruchi.jayaswal@sitpune.edu.in

Abstract

A new coronavirus has caused a pandemic crisis around the globe. According to the WHO, this is an infectious illness that spreads from person to person. Therefore, the only way to avoid this infection is to take precautions. Wearing a mask is the most critical COVID-19 protection method because it prevents the virus from spreading from an infected person to a healthy one. This study reflects a deep learning method to create a system for detecting Face Masks. The paper proposes a unique FMDRT (Face Mask Dataset in Real-Time) dataset to determine whether a person is wearing a mask or not. The RFMD and Face Mask datasets are also taken from the internet to evaluate the performance of the proposed method. The CLAHE preprocessing method is employed to enhance the image quality, then resizing and Image augmentation techniques are used to convert it into a standard format and increase the size of the dataset, respectively. The pretrained Caffe face detector model is used to detect the faces, and then the lightweight transfer learning-based Xception model is applied for the feature extraction process. This paper recommended a novel model that is, CL-SSDXcept to distinguish the Face Mask or no mask images. However, accession with the MobileNetV2, VGG16, VGG19, and InceptionV3 models with different hyperparameter settings has been tested on the FMDRT dataset. We have also compared the results of the synthesized dataset FMDRT to the existing Face Mask datasets. The experimental results attained 98% test accuracy on the suggested dataset 'FMDRT' using the CL-SSDXcept method. The empirical findings have been reported at 50 iterations with tuned hyperparameter values with an average accuracy 98% and a loss of 0.05.

KEYWORDS

3D-face masks, CL-SSDXcept, COVID-19, DNN models, face mask detection, hyperparameters, optimizers

1 | INTRODUCTION

A new coronavirus has become a pandemic, impacting several nations throughout the world. A coronavirus has impacted more than 216 nations, according to a World Health Organization[1,2] assessment. It is also known as Coronavirus, Novel Coronavirus, Corona, and COVID-19. This viral virus was first seen in Wuhan, China, in December 2019, and has since caused an ongoing outbreak. As of 2 June 2022, approximately 527,603,107 cases have been recorded globally, with more than 60 million fatalities, according to a WHO study.[1] COVID-19 is a contagious disease that leads to an infectious sickness. Cough-cold, exhaustion, shortness of breath, sore throat, fever, severe headache, nasal congestion, and other indications are rampant.

There are three phases of the coronavirus that can be transmitted between individuals. This virus is typically communicated between persons during close contact in the early stage, primarily by a droplet formed by sneezing, coughing, or talking. The probabilities of this virus spreading are enhanced in the second stage. Droplets fall to the ground, and individuals can become sick by touching contaminated surfaces and contacting their faces. It becomes a community spread in the third stage. It takes 1–14 days to diagnose a viral symptom. If a person comes into touch with others and the infected person does not display symptoms in the first few days, the virus might spread to the others. The virus can be transmitted if people do not have recommended precautions. The main source to mitigate the virus from one place to another is traveling. If an unhealthy person travels from one place to another, then the virus also spreads to a non-contaminated zone.

However, there are some vaccines have been developed to protect against this virus but these vaccines are not 100% effective therefore, WHO has advised specific preventive measures to lower the risk of this lethal virus.[3] Some preventative practices can assist in preventing infection. Face cover via a mask (especially while interacting with somebody or in a crowded place), frequently washing and sanitizing hands, trying not to touch the face over and over, and social distance all are required to limit the transmission of infections. One more security safeguard executed in India is the ArgoyaSetu App, which cautions persons ahead of time. Assuming healthy people interact with a tainted person or inside a contaminated individual's span, the app gives an alert so having this application on their mobile phone is compulsory. Figure 1 portrays the worldwide situation regarding the number of cases expanding when no security measures are taken while diminishing when safeguards or wellbeing measures are embraced.

After the lockdown time, WHO encourages people to wear a cover and maintain social distance in public places.[2] Coronavirus is not transmitted to another person when wearing a mask; however, those who do not use a mask suffer from cold and cough symptoms. Additionally, many service providers allow consumers to wear Face Masks to access and use their services.

AI technology becomes relevant when a machine or system can do tasks that involve the intelligence of a human. Deep Learning is a subfield of machine learning in which the computer learns from huge volumes of data by behaving like a human brain. Facemask detection application has become a vital computer vision job for assisting society, and basic research has been conducted in this area. Facemask detection approaches based on deep learning will be discussed in this study. This application covers the identification of if people are covering their face with a mask or not and indicates the position of face in boundary box.[3] It is done in real-time on the Face Mask Dataset (FMDRT). Only a few datasets are now existing on the internet. Therefore, we have built a novel Facemask dataset 'FMDRT' due to the scarcity of datasets. Facemask detection uses deep learning techniques to give high accuracy in real-time. In this study, various deep learning models will be tested on FMDRT.

Facemask detection has emerged as a new real-time challenge for the authors. Nevertheless, there are limited images available of Face Masks on the internet. Face detection is a crucial stage in the Face Mask Detection system that can predict faces in live video. Additionally, some pseudo-codes and datasets are designed to address this issue in real-time. A reliable dataset and an effective algorithm are essential to building a robust Face Mask detection application in a pandemic situation. The purpose of this study is to cover these two issues. Firstly, a new dataset, FMDRT, is developed and offered a new technique, CL-SSDXcept, which is a very efficient technique to train the model on a novel dataset.

The critical contribution of this study is that it makes novel research dataset and techniques different from past work in the Face Mask detection field. As many new researches have been done in this field, but some concepts remain to cover in this application which turned to make the novelty of this work. The main aim of the article is discussed below.

- Due to the scarcity of Face Mask datasets, we have introduced a unique Face Mask dataset, that is, FMDRT (Face Mask Dataset in Real-Time) includes various Face Masks such as natural-looking human appearance 3D_printed masks, 2D masks, transparent masks, and clown printed masks. A total of 851 images resides in this dataset, of which 551 are with Face Mask images, and the remaining 300 are without Face Mask images.

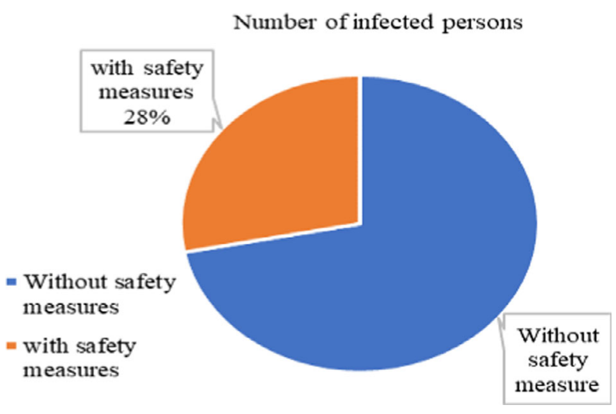


FIGURE 1 Possible spread of COVID-19 with or without precautions globally[35]

- These images have been captured at various occlusion and illumination effects, taking various postures of the human faces at different orientations.
- A CL-SSDXcept method is proposed to detect Face Masks in real that bind the pre-processing technique, face detection SSD model, and Xception model for classification purpose.
- Rather than the proposed model, the system also tested different models with tuned hyperparameter settings and evaluated the results.
- At last, the recommended model is especially compared with the past research model on FMDRT and existing Face Mask datasets and assesses the impressive results.

The organization of the paper is partitioned into six parts. Section 2: The related study will explain and review it. Section 3 will detail the proposed Face Mask detection methodology. Section 4: Material required will summarize the proposed Face Mask datasets. Section 5: Experimental Configuration, the evaluation metrics, outcomes, and result analysis will be covered, and finally, Section 6 will go through the conclusion.

2 | RELATED WORKS

Mingjie Jiang et al.[4] proposed "RetinaFaceMask," research for distinguishing facial coverings that is proficient and precise. He presents a special context attention module methodology for perceiving facial coverings. He took advantage of the publicly available Face Mask dataset. He also gave a cross-class object removal method that eliminates low confidence forecasts and the union's excessive intersection rate. This method employed MobileNet and ResNet models to identify mask or no mask. The recall and accuracy of this procedure to recognize cover using MobileNet + RetinaFaceMask are 89.1%, 82.3%, and 82.3%, respectively, and 94.5% and 93.4% while utilizing RetinaFaceMask and ResNet.

The three Face Mask datasets were proposed by Zhongyuan Wang et al.[5] freely available. The dataset are easily accessible to everyone and published on Github. The (MFDD) Masked-Face-Detection Dataset is divided into two portions. Some of the samples are from relevant studies, while others are taken from the Internet. MFDD includes 24,771 mask pictures. The next dataset is the (RMFRD) Real-Mask-Face-Recognition Dataset. It has shots of 525 individuals wearing masks and 90,000 images of the same 525 people wearing no masks. This is the massive real-world mask versus no mask dataset. Lastly, the Stimulated-Mask-Face-Recognition Dataset (SMFRD) was created artificially by putting masks on existing face datasets.

The Masked Faces dataset, which comprises 30,811 web pictures and 35,806 Face Masked images, was used by Shiming Ge et al.[6] to solve the face detection problem. Various degrees of occlusion present in images and with different orientations of face, images are present in the collection. For Face Mask identification, LLE-CNNs with three major modules are also demonstrated. Two pre-trained CNNs are used in the first module to extract facial attributes from input images and consider these as high-dimensional descriptor. The Local Linear Embedding (LLE) approach is then used to convert these descriptors into similarity-based descriptors. Lastly, the verification module detects the candidate's facial regions, which integrates classification and regression tasks into a single CNN. According to the MAFA dataset, the recommended approach exceeds states-of-the-arts at 15.6%.

Ashu Kumar et al.[7] surveyed various face detection techniques. Several challenges and applications are also discussed in this paper. Face detection approaches are described and include Feature-based methods, Image-based approaches, and presented as their subtypes. He described a comparative study of both systems. It also described several datasets for face detection and available facial recognition APIs.

Jiankang Deng et al.[8] recommended a powerful RetinaFace that performs pixel-wise face localization and uses extra-supervised as well self-supervised multitask learning. The study is processed in five modules. In the first module, he manually labels five facial landmarks on the WIDER Face dataset and observes the improvement in hard face identification when an additional supervision signal is used. The second module includes a self-supervised mesh decoder branch for assessing 3D shape face information at the pixel level. RetinaFace improved the state-of-the-art average accuracy on the WIDER Face hard test set by 1.1% in the third phase. In the fourth phase, RetinaFace enables cutting-edge technologies to increase overall face verification outcomes (TAR and FAR are equal to 89.59 and 1e-6). At last, utilizing lightweight backbone organizations, RetinaFace might run progressively on a solitary CPU center for only a VGA goal picture in the end module.

Faen Zhang et al.[9] proposed AInnoFace, an efficient method for face identification. The process starts with the first stage, RetinaNet, and is then implanted with several strategies to achieve high performance. For regression, he employed the Intersection over Union loss function, two-step classification, and regression to detect Face, and image augmentation based on the data anchor sampling for training. He employed the max-out operation and the multiscale testing method for categorization. Consequently, our method enhances face detection performance on the WIDER face dataset.

Md. Rafiuzzaman Bhuiyan et al.[10] recommended a Face Mask detection approach using the object-based detection YOLOv3 model. The work starts from the Data acquisition technique, and the images are annotated using the web LabelImg tool, then extracting the features of images using the YOLOv3 model. Lastly, they classify the images into a mask or no mask. The experimentation is performed on 600 web-collected images where 300 images are mask and the remaining no mask images. The accuracy obtained in real-time was 96% after 4000 iterations.

Preeti Nagarath et al.[11] suggested a Face Mask detection system by following the MobileNetV2 model to predict the mask or no mask images. The experiment is performed on Face Mask dataset that is collected from web in which 690-mask images and 686- without mask images. The dataset contains simple Face Mask images resides such as printed masks and surgical masks but there is no other different types of Face Masks. The accuracy obtained on this dataset is 0.92 and F1-score achieved 0.93.

3 | PROPOSED METHODOLOGY

The presented method in this part introduced an approach known as CL-SSDXcept. This section discusses how to spot a Face Mask using deep learning models. The system predicts whether persons have correctly donned masks using the suggested technique; the first stage (training phase) would be to train a model using a recommended FMDRT dataset and an existing dataset. Sections 4.1 and 4.2 go into some depth regarding the FMDRT and existing Face Mask datasets. After fitting the network as a classifier, a model for detecting faces is required to recognize faces in video frames so that the CL-SSDXcept model can predict whether the person having a mask or no mask. The work aims to improve the accuracy of a Face Mask detection system in minimum computation time while being consistent across all datasets. A workflow of the proposed methodology employed in the study is depicted in Figure 2. Several models such as MobileNetV2,[12] VGG16,[13] VGG19,[13] InceptionV3[14] are utilized to compare with the suggested technique at the benchmark, and potential hyperparameters are altered to assess the outcomes.

The suggested CL-SSDXcept approach is divided into two stages, as indicated below. The Face Mask datasets are loaded as input for the training phase in stage 1 of the training method, as illustrated below. The image augmentation approach is used to increase accuracy. Following that, image data is partitioned into training and validation sets. After that, use a transfer learning-based CNN model[15] to retrieve essential features from Face Mask photos. An Xception model trains the features and classifies them according to the user's needs.

Furthermore, the Xception model is FineTuned feature extraction based on the model's needs, accuracy, and the computation time acquired for the training and validation sets. In Phase 1 and Phase 2, the algorithm is tested as indicated below. The trained model was then deployed on both datasets, and real-time images were captured using a camera during the testing phase. Using the border-box, apply the Single Shot Detector (SSD) model for detecting the faces inside the video frame. The Face Mask Detector trained model is then used to identify faces as 'with mask' or 'no mask' by making the border-box. The training and testing stages of the Face Mask detection system are demonstrated step by step.

The flowchart of Face Mask detection is described stepwise for the training and testing phase.

Phase 1. A training stage

Input data: Acquire Face Mask image dataset (FMDRT); each image is captured at 2 to 6 ft away from the camera, Image size ($a \times b \times c$).
Output data: Facemask detector model (.h5 format).
Step 1: Grab the Face Mask dataset from the disk drive.
Step 2: Pre-process using the CLAHE technique and resize the images.
Step 3: Apply the image augmentation technique and partition the images into training and validation batches using the function.
Step 4: Use CNN Xception model (size = [299,299,3]) to extract the features.
Step 5: Set and change the hyperparameters of the proposed model.
Step 6: Apply ADAM optimizer to improve the features set.
Step 7: Acquired and saved the Face Mask detector (.h5) model, and checked model precision and model generation time.

Phase 2. A testing stage

Input: Load and capture pictures from test dataset and real-time image/video stream, load face detector, and Face Mask detector model. Image size ($a \times b \times c$).
Output: Determine people are wearing "mask" or "no mask".
Step 1: Attaining images for the test set and captured video frames from a webcam in real-time.
Step 2: Pre-process the image (apply CLAHE and Resizing function).
Step 3: Apply a predefined face detector model to locate faces in an image that is, the Caffe prototype SSD model.
    If person's face is a detector in real-time frames.
        Crop and Resize the face by marking the border box coordinates.

Step 4: Employ obtained Face Mask detector (.h5) model and display truthiness;
Get the prediction in testing frames.
If a person covers face via mask,
then display on the border box 'With mask'.
Otherwise,
then display on border-box 'No mask'.
Step 5: When "q" is pressed, the real-time video stream is terminated.

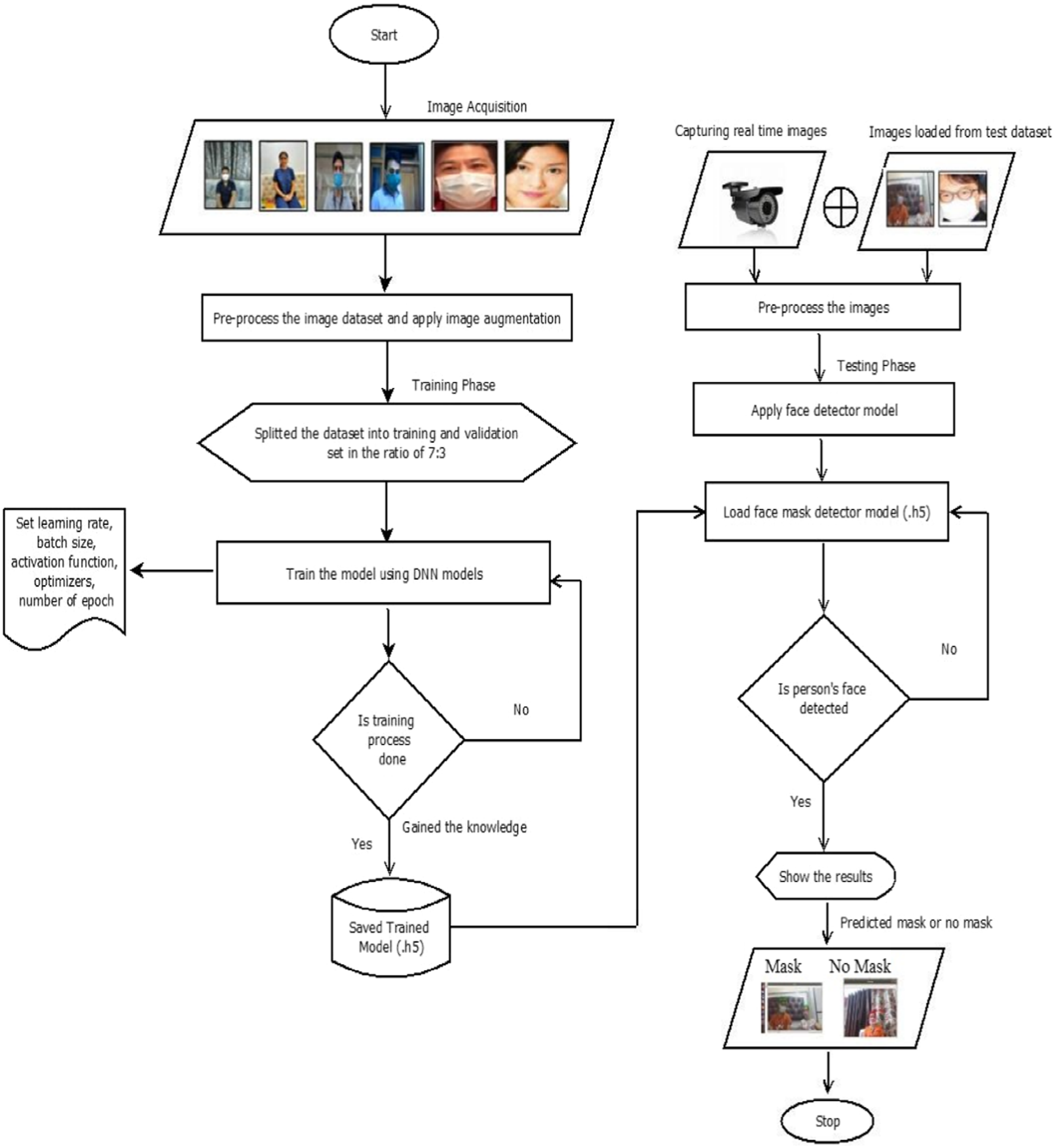


FIGURE 2 The workflow of the proposed methodology

3.1 | Image pre-processing and augmentation of Face Mask dataset

The primary step of the methodology is pre-processing the image database, which involves image acquisition, that is, capturing the number of Face Mask images. Since fewer Face Mask datasets are publically accessible, we gathered images with and no Face Masks. A brief description of the Face

Mask datasets will be elaborated in Sections 4.1 and 4.2 later. Because the RFMD. Dataset is relatively huge, only 3000 pictures were chosen for the experiment. Divide the dataset into train and test sets of images using a function. All images are preprocessed and resized. Images are resized into standard resolution as per the model's requirement. The images are transformed to a NumPy array following picture resizing for simpler and quicker computation.

3.1.1 | Contrast-limited-adaptive-histogram-equalization approach

CLAHE[16] (Contrast-Limited-Adaptive-Histogram-Equalization) is a sort of adaptive histogram equalization that limits contrasting augmentation to minimize amplification of noise. The restricted contrastive method must be applied for each neighborhood through which a transformation function is produced. Rather than obtaining the full image, over amplification is avoided by separating the data into little data pieces called tiles, that are then contrasted. The tiles are then put together to form a complete image. It works with grayscale as well colored RGB images. Figure 3 presents an example image of a Face Mask created using the CLAHE technique.

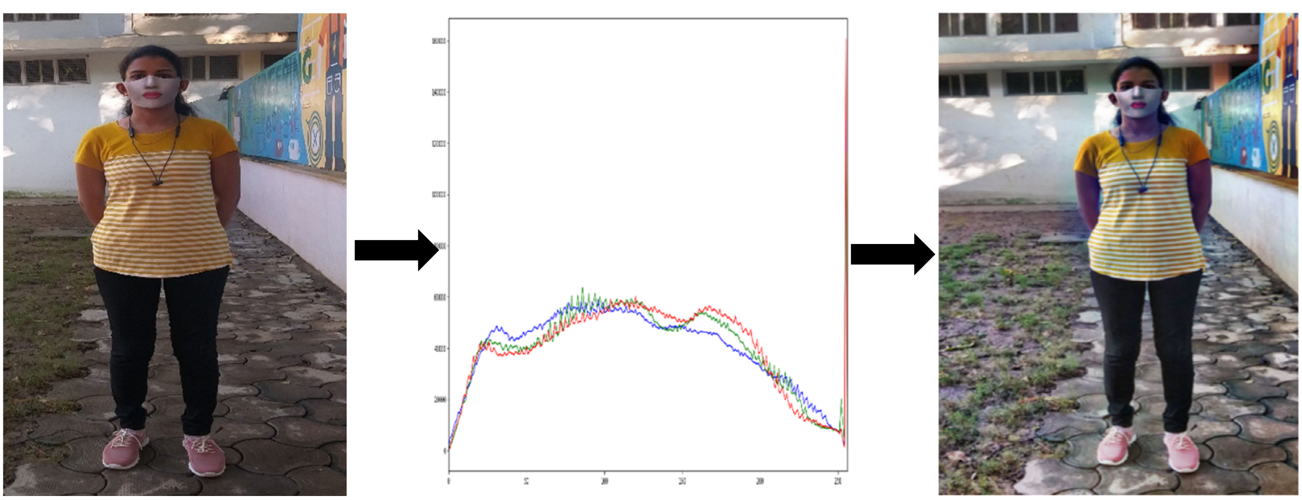
3.1.2 | Data augmentation approach

Data augmentation is also utilized to improve the training model's accuracy. To train deep learning models for improved accuracy, a large amount of data is necessary. Following image augmentation of the dataset, the CL-SSDXcept model is trained in this suggested framework. The augmentation method is particularly beneficial for enhancing performance since the FMDRT dataset is inadequate for training. Utilization of augmentation techniques like rotation, flipping, shifting, shearing, zooming, and brightness to scale the size of our recommended dataset. This method is utilized to produce superior outcomes in both datasets. The image DataGeneration function is aid to supplement data/images, and it returns image batches for training and validation. The sample files are shown in Figure 5 after applying image augmentation on particular image. Rotation, Flipping, Shearing, and Brightness augmentation effect are shown in Figure 5.

Syntax to import the augmentation files from pre-processing layer: From tensorflow.keras.preprocessing.image import Image DataGenerator.

The six augmentation techniques are used to increase the size of dataset and the syntax is as follows:

```
aug=ImageDataGenerator(.
rotation_range=30,
```



(A) Original Image (B) Histogram of image (C) An image after CLAHE technique

FIGURE 3 Impact on image after CLAHE technique

```
zoom_range=0.15,
width_shift_range=0.2,
height_shift_range=0.2,
shear_range=0.15,
horizontal_flip=True,
vertical_flip=True,
brightness_range=[0.5, 1.7].
fill_mode="nearest").
```

Benefits to use the augmentation layers in the Face Mask detection application are shown in Figure 4.

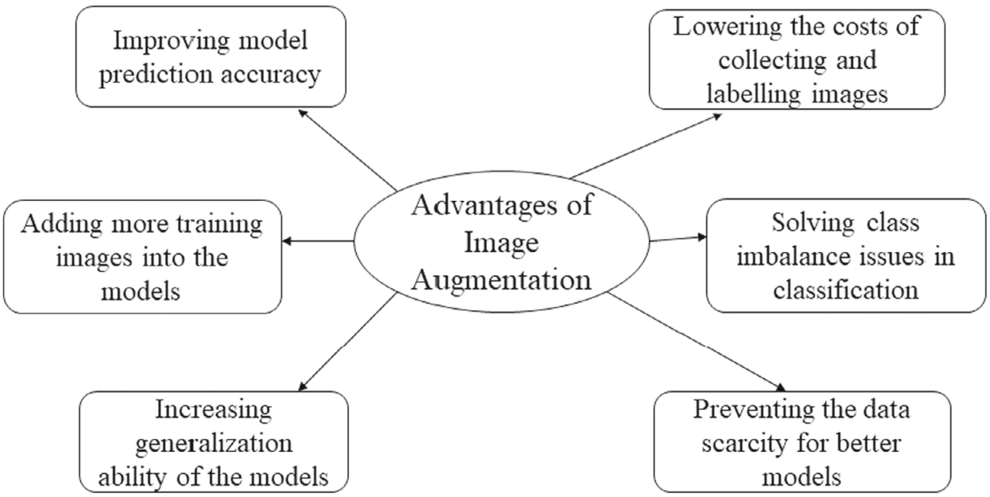


FIGURE 4 Benefits of image augmentation

3.2 | Face detector (SSD) caffe detector model

Face detection is one of the crucial step of the methodology. The first step of the Face Mask Detection framework is to locate the faces in test Face Mask images or detect faces in real-time frames. At the time of the testing phase, when a person's face comes in face of the CCTV or webcam, it becomes necessary to detect the face first and then check the person is wearing a mask or without a mask. So, the caffe model is used to detect the face. This model is based on single shot-multibox detector (SSD). It follows ResNet- 10 architecture as its backbone. Caffe model[11] comes in a deep neural network module that was introduced post OpenCV 3.3. Two files are loaded to detect the face (i) Caffe model (ii) prototxt files. After catching the face using the SSD model by writing "cv2.dnn.readNet ("path/to/prototxtfile","path/to/caffe-weights")", the faces are detected by making boundary boxes of the face. By using the caffe model, in real-time frames, faces are detected even in the face's distinguish orientations, that is, right, left, bottom and top. We have two models for CL-SSDXcept proposed work, the Caffe model and the Xception model, for classifying the mask or no mask image.

3.3 | Feature extraction and classification

3.3.1 | Classification of images using transfer learning-based Xception architecture

The feature extraction and classification of the image dataset are done by transfer learning-based fine-tuned[17] Xception model. The model's input image size is 299×299 resolution. It builds on the InceptionV3 design, using depth-wise separable convolutions to replace standard Inception modules. Among the ResNet-152, VGG-16, and InceptionV3 models, the Xception model has the highest accuracy, with top-1 and top-5 accuracy of 0.79 and 0.945, respectively. Xception is a modified architecture that is essentially comprised of two components.[18] Convolution block shortcuts in the ResNet style are the first, whereas depth-wise separable convolution is the second. There are 71 layers in this network. As illustrated in Figure 6,

FIGURE 5 Sample files after image augmentations

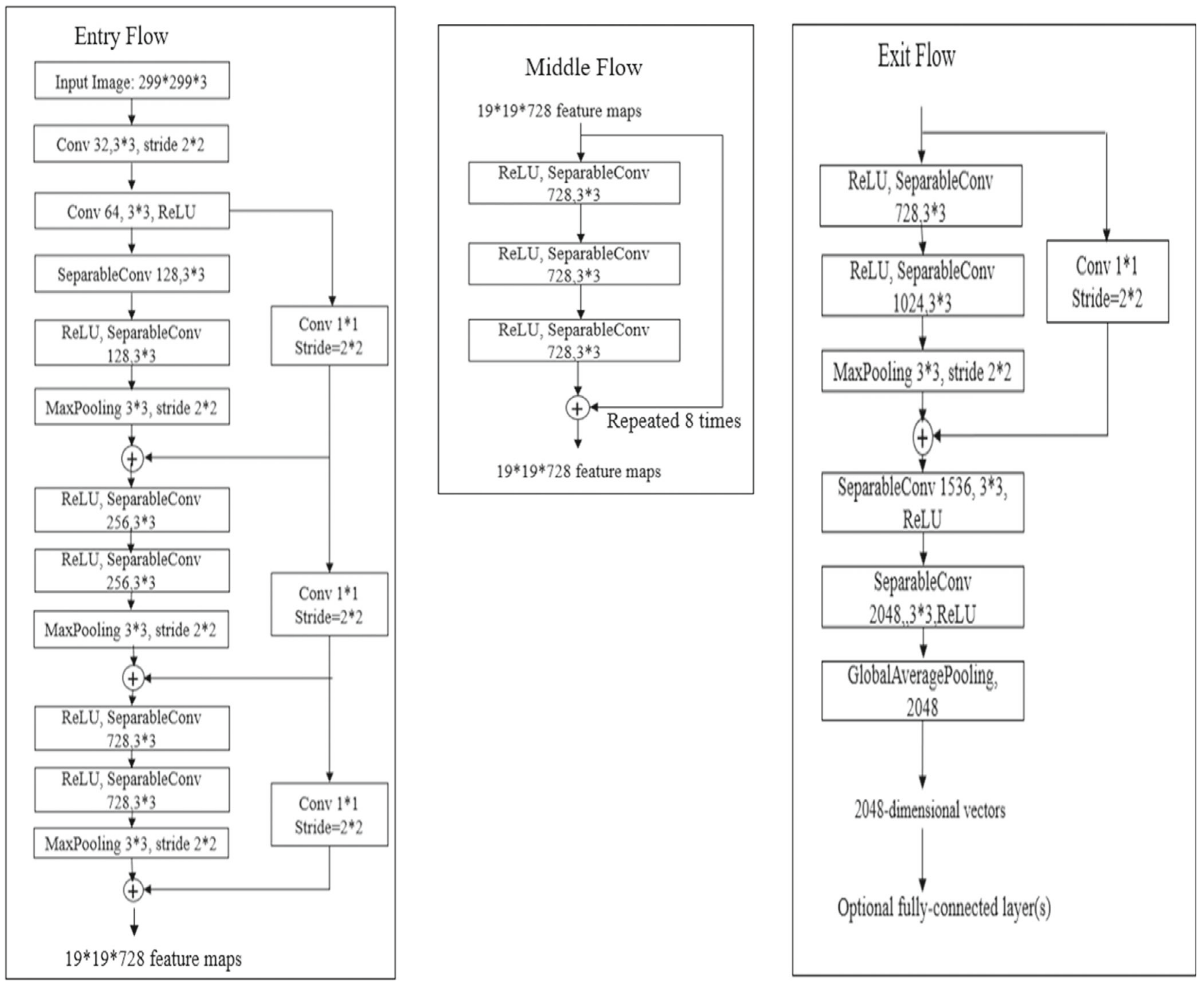


FIGURE 6 Architecture of Xception model

the Xception design is divided into three key segments: "Entry Flow, Middle Flow, and Exit Flow". There are 36 convolution layers organized into 14 modules in the framework. All other modules are ringed by straight residual connections, except the first and last. Collet[18] described the Xception[19] architecture as a straight stack of depth wise-separable convolutions with feedback connections when trained on the ImageNet dataset.

3.3.2 | Transfer learning approach on Xception model

A model may be trained with the aid of a transfer learning technique by simply switching from one usage to another.[20,21] There are several parameters in the Xception model that must be learned. These parameters are frequently selected at random at the initial training phase, producing a very significant upfront error for the network, which can lead to poor convergence and over-fitting difficulties. A supervised pre-training approach of transfer learning based on feature selection is developed to deal with this challenge. The goal is to retrieve common visual features in the source, target domains, and then execute knowledge transfer using these feature extractions. The Source and Target Domains of Transfer Learning are defined as follows:

$$\mathrm{TL(s)} = \{\mathrm{x}, \mathrm{PD(x)}\} \tag{1}$$

$$\mathrm{TL(t)} = \{\mathrm{x}, \mathrm{PD(x)}\} \tag{2}$$

where TL(s) denotes the source region and TL(t) denotes the target region; x and PD(x) denote the minimum probability distribution function for the feature vector and map in single region. The source task TL (s) in the source region is Imagenet recognition. In contrast, the target task TK (t) in the target region is Mask or no mask face image classification. The feature information of source region is transfer to the target region. The pretrained network from the origin region is used to build the model parameters in the target job. Transfer learning is a method of applying previously acquired knowledge to new but similar situations. The Xception model's last layer is eliminated, allowing the model to fine-tune to the model's needs. The model pertains more beneficial features than other models on the proposed dataset.

3.3.3 | Optimizers

An optimizer is now essential in training a deep learning model to reduce error rates. The optimization approach aids in improving the performance and speed of model training. Optimizers are a new class with more information for training a model. There are various optimizers available, including SGD,[22] AdaGrad,[23] RMSProp,[23] ADAM,[24] ADADELTA,[25] Nesterov,[26] and Ftrl.[27] Each has advantages and disadvantages. The Adaptive Moment Estimation optimizer is employed with the proposed model, and after evaluating the results, the outcomes are collet and discussed in Table 2.

Adaptive-Moment-Estimation (ADAM): It is now widespread and effective in deep learning models and is a hybrid of the RMDProp and Momentum methods. The procedure is similar to RMSProp, unless a sleek version of the gradient is employed despite of raw stochastic gradients. Adam adds a bias correction as well. It is simple to implement and uses less memory.[24] Equations (3), (4), (5), and (6) illustrate how to update the gradient and contain a bias correction mechanism for each parameter m_{j}.[28]

$$l_t = \theta_1 * l_{t-1} - (1-\theta_1) \times g_d \tag{3}$$

$$s_g = \theta_2 \times s_{g-1} - (1-\theta_2) \times g_d^2 \tag{4}$$

$$\theta = -\alpha \times \frac{l_t}{\sqrt{s_{g+\varepsilon}}} \times g_d \tag{5}$$

$$m_{t+1} = m_t + \Delta m_t \tag{6}$$

Here, l_t denotes the exponential average of gradients along m_{j}, g_{d} equivalent to gradient at time t along m_{j}, α is the learning rate (LR). θ_1 and θ_2 are the hyperparameters. The exponential average of square of gradient at m_{j} is s_{g}.

Evaluate the gradients exponential average and also the squares of the gradient Equations (3) and (4) for each parameter values. In Equation (5), multiply the LR by the averaging of the gradient and divide by the Root Mean Square of the exponential average of square of gradients to get the learning step. After that, the adjustment is applied. The hyperparameter values θ_1 is commonly set to 0.9 and θ_2 set to 0.99. Generally, ε is set to 1e-10.

4 | MATERIAL REQUIRED

4.1 | Synthesized Face Mask dataset: Face mask dataset in real time

A novel Face Mask dataset has been proposed to detect the Face Mask in public places. There are two broad categories of faces available in this dataset that is, with masks and without masks. The Face Mask images are different from other Face Mask images because it contain several kinds of masks such as 3d printed, skin color, human facial appearance, natural-looking Face Mask appearance, male–female appearance mask, and transparent joker or clown printed Face Masks. Without Face Masks, images of males and females are available in this dataset. All the images are captured through a webcam and mobile phone placed on a tripod, and the distance between camera and person is 2 to 7 ft. The proposed dataset contained 851 images, of which 551–with Face Masks and 300–were without Face Masks. Samples of the dataset are shown below in Figures 7 and 8. Following are some characteristics of this dataset that make it different from other available Face Mask datasets.

- FMDRT dataset is suggested because it includes 3D printed masks, clown printed, natural human-looking appearance masks, and transparent Face Masks. There is no such type of dataset available on the internet.
- Pictures are collected at different locations such as colleges, laboratories, homes, parks, and grounds with varied lighting effects under sunlight, tube lights, and led lights.
- Images are captured with changing illumination, occlusion, lightening effect, and varied postures to check the robustness of the suggested technique.
- Different angles of faces such as top, bottom, left, and right has been taken wearing or without a mask.
- Real-time images are captured to make the dataset and consider one and more than one person images in a single frame.

Sample images of Face Masks such as 3d printed, skin color, human appearance, and clown masks of FMDRT dataset are shown in Figure 7.

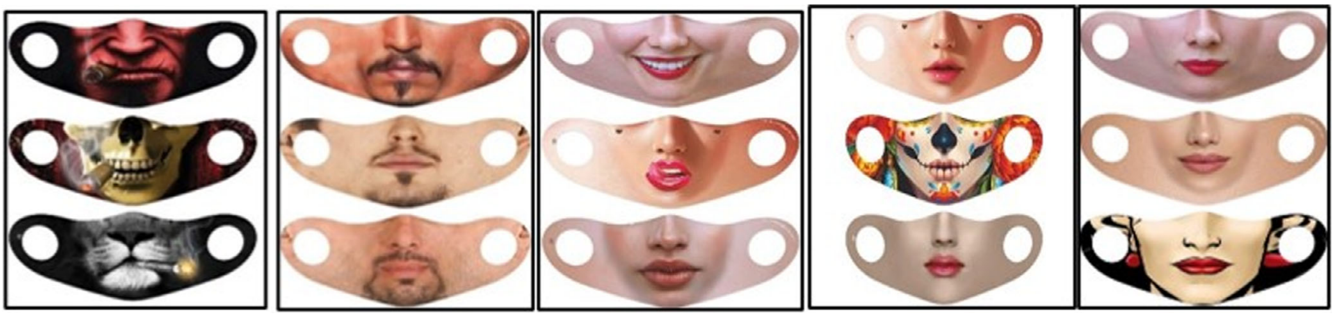


FIGURE 7 Sample images of Face Masks of FMDRT dataset

FIGURE 8 Shows the sample images of the FMDRT dataset

FIGURE 9 Images of real world masked face dataset

FIGURE 10 Samples of Face Mask dataset

4.2 | Existing Face Mask datasets

4.2.1 | Real world masked face dataset

The following dataset, RFMD,[2] was obtained from the web and resides 2 kinds of face images, masked and without masks. It had 5000 masked faces of 525 individuals and 90,000 unmasked faces of people. These images were acquired at railway stations and collected from the web. Yet, we only took 3000 pictures, 1500 of which featured mask faces and 1500 without masks. Figure 9 represented the sample images of RFMD.

4.2.2 | Face mask dataset

The Face Mask dataset[15] contains 7959 pictures of people with and without masks on their faces. On the other hand, this dataset is a combination of the Wider Face and Masked Faces datasets. Wider Face is a collection of 32,203 images of 939,703 ordinary individuals in diverse lighting, occlusions, and poses. MAFA includes 34,806 masked images and 30,811 normal face pictures. Instead of masks, some of the faces are disguised behind hands or other things. The collection comprises noisy and corrupted images. Figure 10 presented the sample images of Face Mask Dataset.

5 | EXPERIMENTAL DESIGN, EVALUATION METRICS, EVALUATIONS, OUTCOMES, AND ANALYSIS

This section discusses the experimental design, evaluation metrics, evaluations, outcomes, and result analysis. Different models with variable hyperparameter values are examined to generate the model's accuracy. Hyperparameter settings are fine-tuned to evaluate system performance. A hyperparameter is a parameter that may be optimized before the learning process begins. The configuration of hyperparameter values is kind since it changes with each new application.[30] Hyperparameter tuning is selecting a set of optimal hyperparameters for a machine learning or CNN methods.[31] To evaluate the system's performance, this study considers Learning Rate (α), Batch Size (m), and Optimizers. Depending on the hyperparameter values, different results are obtained.

5.1 | Experiment design

The hardware equipment with 16GB RAM, an I7 core CPU, and a 4GB NVIDIA GPU processor is used for experimental purposes. The Spyder platform is used for implementation, and the models are then trained. We have partitioned the dataset into a train and validation set with 70% and 30% pictures, respectively, using split function in python and acquainted with certain models. The entire experiment is performed on the GPU environment so that computation time is enhanced tremendously compared to the non-GPU environment.

5.2 | Evaluation metrics

As metrics, this system used Precision (PR), Recall (RC), Accuracy (AC), and Loss or Error Rate, which are expressed as follows.[32,33]

5.2.1 | Precision is defined as estimating correctness and evaluating the ratio of positive consequences from results properly identified in the test window

$$PR = \frac{\mathrm{TPC}}{\mathrm{TPC} + \mathrm{FPC}} \tag{7}$$

5.2.2 | Recall defined the proportion of appropriate query responses in a dataset to the total amount of images

$$\mathrm{RC} = \frac{\mathrm{TPC}}{\mathrm{TPC} + \mathrm{FNC}} \tag{8}$$

5.2.3 | Accuracy is explained as the proportion of successfully predicted classes to total tested categories in an image dataset

$$\mathrm{Accuracy} = \frac{\mathrm{TPC} + \mathrm{TNC}}{\mathrm{TPC} + \mathrm{TNC} + \mathrm{FPC} + \mathrm{FNC}} \tag{9}$$

Here, TPC = true positive class; TNC = true negative class; FPC = false positive class, and FNC = false negative class.

5.2.4 | Sigmoid cross-entropy loss is another name for binary cross-entropy loss. Then, employ it when we have a binary classification problem (C = 2)

$$\mathrm{CL} = -\sum_{i=1}^{c=2} t_i \log(f(s_i)) = -t_1 \log(f(s_1)) - (1 - t_1)\log(1 - f(s_1)) \tag{10}$$

Here, CL = cross entropy loss,

$t_1 = 1$; The class $C_1 = C_i$ is present,

s_i = Gradient with respect to each score.

5.3 | Experimental evaluations

Table 1 demonstrates how the model's learning rate and batch size parameter affect truthiness and computation time. The 50 epochs or iteration is used to train the model, and the total number of Dense Layers (DL) is 256; the Activation Function (AF) = Sigmoid, (Binary classification) and the Adam optimizer are used. The critical parameter to adjust in CNN models is the learning rate; thus, the effect of changing the learning rate is shown

TABLE 1 Adjusting the learning_rate and batch_size has an effected the accuracy and time elapsed

		Accuracy (%) and Time (ms)								
		Learning_rate (α) Batch size (m)								
		0.0001	0.0001	0.0001	0.001	0.001	0.001	0.01	0.01	0.01
S. No.	Models	16	32	64	16	32	64	16	32	64
1	MobilenetV2	92%, 500.69 ms	93% 502.69 ms	94% 499.71 ms	93% 505.77 ms	95% 504.52 ms	94% 503.4 ms	93% 505.22 ms	94% 499.76 ms	93% 502.87 ms
2	VGG16	70% 630.21 ms	76% 625.22 ms	68% 620.68 ms	85% 632.69 ms	86% 628.21 ms	80% 621.18 ms	88% 629.49 ms	80% 626.86 ms	80% 622.18 ms
3	VGG19	74% 735.63 ms	71% 732 ms	64% 734.0 ms	88% 735.7 ms	82% 730.65 ms	78% 729.462 ms	81% 731.80 ms	80% 733.92 ms	81% 730.2 ms
4	InceptionV3	95% 1121.58 ms	94% 1175.89 ms	94% 1130.13 ms	95% 1128.76 ms	97% 1120.6 ms	94% 1124.93 ms	96% 1201.27 ms	95% 1120.35 ms	95% 1158.09 ms
5	CL-SSDXcept	96% 1096.77 ms	95% 1090.66 ms	95% 1051.49 ms	98% 1068.05 ms	98% 1050.29 ms	98% 1051.591 ms	98% 1062.61 ms	97% 1066.43 ms	97% 1068.62 ms

in Table 1. On the FMDRT dataset, the CL-SSDXcept model has higher average accuracy than all models. The comparative analysis of other models proposed on our generated dataset, FMDRT, is presented in Table 1.

Figures 11, 12, 13, and 14 depicted the training, validation accuracies, and losses of different models. In addition to the CL-SSDXcept model, the InceptionV3, Mobilenet, and VGG16, VGG19 models are also used to train the dataset, measure accuracy, and lead the CL-SSDXcept model to perform better than other models. VGG16 and VGG19 have not performed well in terms of train and validation accuracy. We may deduce from the findings that the suggested model and InceptionV3 give superior accuracy. However, CL-SSDXcept generates the best accuracy with less computing time than the InceptionV3 model. Hence, we choose the CL-SSDXcept model for our suggested study for Face Mask identification. Moreover, it can infer from Table 1 that the learning rate at 0.001 and batch size at 32 produce good results; therefore, for further findings, we will set these hyperparameters at the same values. Table 2 demonstrates that several optimizers have been used to analyze the performance of the system.

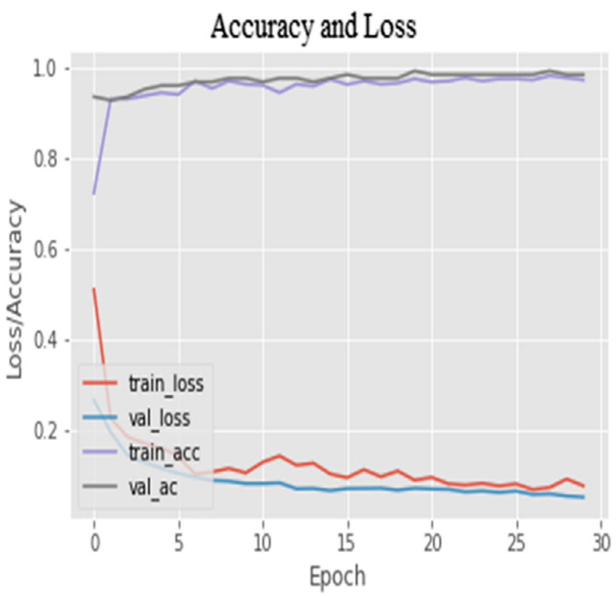


FIGURE 11 CL-SSDXcept model

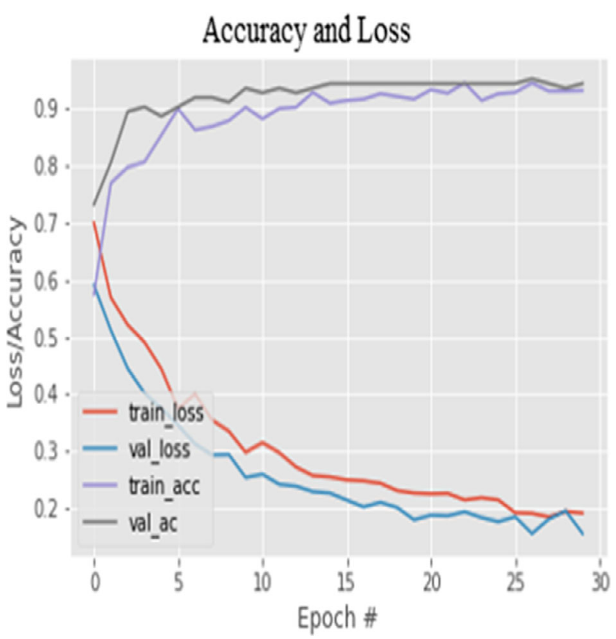


FIGURE 12 InceptionV3 model

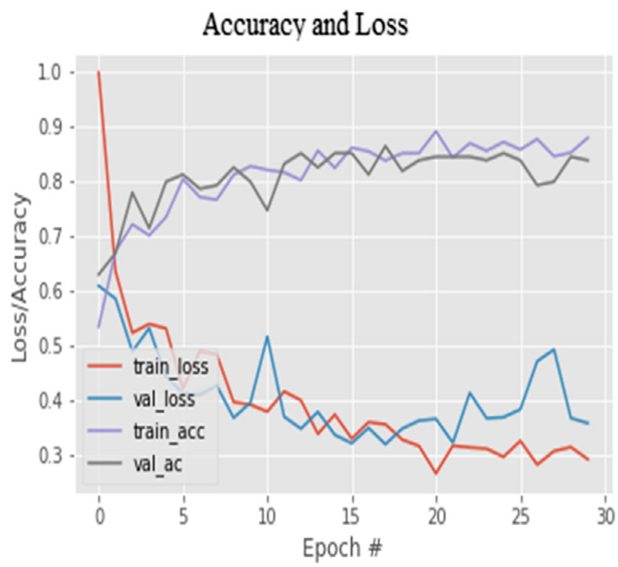


FIGURE 13 MobilenetV2 model

FIGURE 14 VGG19 model

TABLE 2 Adjusting optimizers has an effect on model accuracy and time elapsed

		Accuracy (%) and Time (ms)						
		Numerous optimizers, $n = 50$						
S. No.	Models	SGD	RMSProp	AdaGrad	ADAM	AdaDelta	NADAM	Ftrl
1	MobileNetV2	92%, 507.2 ms	94%, 530.17 ms	91%, 521.74 ms	95% 490.4 ms	54%, 514.08 ms	94%, 588.3 ms	89%, 585.3 ms
2	VGG16	61% 717.5 ms	85%, 709.97 ms	54% 722.55 ms	58% 652.1 ms	52% 735.41 ms	85% 724.91 ms	53% 733.6 ms
3	VGG19	64% 787.7 ms	79% 790.94 ms	57% 792.51 ms	78% 720.46 ms	38% 763 ms	79% 781.8 ms	53% 776.7 ms
4.	InceptionV3	91% 1412 ms	95% 1429 ms	89% 1431.26 ms	97% 1274.5 ms	71% 1426.09 ms	93% 1428.4 ms	53% 1430 ms
5.	CL-SSDXcept	96% 1365.3 ms	92% 1354.60 ms	92% 1350.30 ms	98% 1272.9 ms	70% 1351.66 ms	95% 1354.34 ms	76% 1343.28 ms

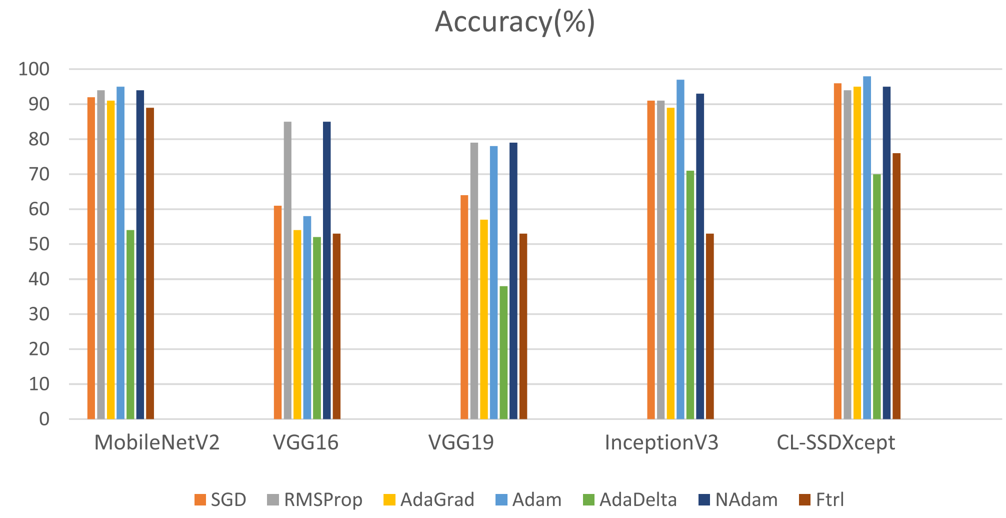


FIGURE 15 Graph depicts the overall comparison of seven optimizers in 5 CNN models.

The graph in Figure 15 shows the results of five different optimizers on different models. It can infer from Table 2 and Figure 14 that ADAM and NADAM perform well compared to respective optimizers on proposed dataset. The FTRL and AdaDelta optimizers do not provide satisfactory outcomes. SGD, AdaGrad, and RMSProp perform better than AdaDelta and Ftrl optimizers. The InceptionV3 model also produces comparable results to the proposed model but takes longer computation time than CL-SsdXcept.

5.4 | Outcomes

Figures 16 and 17 display the outcomes in real-time video that predicts the person wearing a mask or without the mask by marking a boundary box and showing the accuracy on that box. The results have been obtained using the CL-SSDXcept model.

Furthermore, we have performed experiments on the FMDRT, RFMD, and Face Mask datasets and compared their performance, as shown in Table 3. In this case, the learning rate is 1e-4, the batch size is 32, the epochs are 50, the activation function is Sigmoid, the dense layer is 512, and the ADAM optimizer are tuned and test the performance of the system using proposed technique.

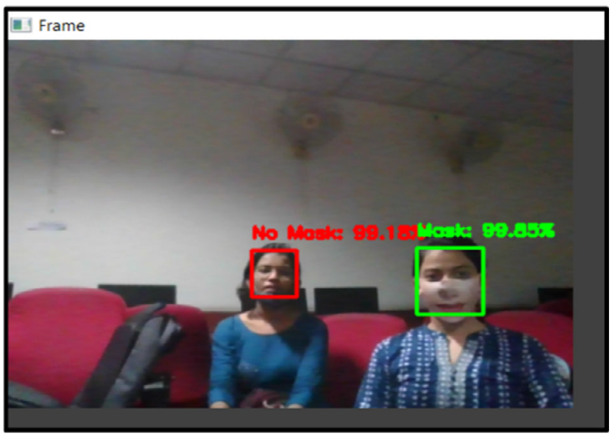


FIGURE 16 Predict with no mask and with mask

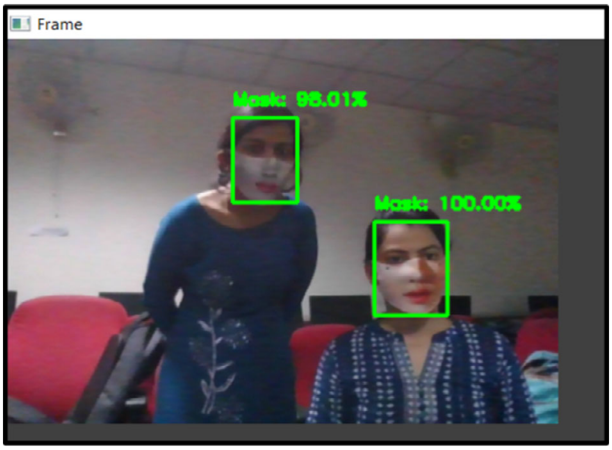


FIGURE 17 Display Face Masks with accuracy and show accuracy

TABLE 3 Train and validation accuracies on synthesized as well existing datasets using the suggested model

S. No.	Dataset name	Total number of images	Suggested model	Train accuracy	Train loss	Validation accuracy at real time
1	**FMDRT (Proposed dataset)**	**851**	**CL-SSDXcept**	**98.77%**	**0.05**	**98.78%**
2	RFMD	3000	**CL-SSDXcept**	98.85%	0.08	97.16%
3.	Face Mask Dataset	7000	**CL-SSDXcept**	96.16%	0.11	95.24%

TABLE 4 Comparison of the proposed model against the state-of-the-art models on FMDRT dataset

Models	Accuracy	Time required (ms)	Total number of parameters	Model size
MobileNetV2	92%	504.52	2 Million	20 MB
VGG16	86%	628.21	14 Million	64 MB
VGG19	82%	730.65	21 Million	83.8 MB
InceptionV3	97%	1120.6	21.9 Million	96 MB
CL-SSDXcept	98%	1050.29	22.3 Million	98 MB

5.5 | Result analysis

Upon analyzing the above experimental results from Table 1, it can infer that the proposed method works well compared with the other models. The CL-SSDXcept model achieved outperforming accuracy on the synthesized datasets by tuning the hyperparameters and can conclude that the learning rate at 0.001 and mini-batch size at 32 produced better results rather than the tuning. It can also be noted in Tables 1 and 2 that the accuracy achieved from the proposed model is approximately the same as the InceptionV3 model, but less training time is required to train the CL-SSDXcept model, and ADAM optimizer works well for this application. Table 4 discusses the state-of-the-art models based on accuracy, computation time, F1 score, the total number of parameters, and model size evaluated on the synthesized dataset. The study's goal is to attain the highest accuracy with less computation time and space. The precedence of all taken parameters is used to choose the best model that is, Accuracy > Time Required > Model Size > Number of parameters.

Consequently, MobileNetv2 is the most miniature model in terms of various parameters, elapsed time, and model size, but its accuracy is poor; thus, it will be ignored as per the precedence. Furthermore, the VGG16 and VGG19 models produce fewer (trainable and non-trainable) parameters but with worse accuracy. The CL-SSDXcept model had good accuracy among all the models along with the number of parameters, training time elapsed, and the model size is less than in the InceptionV3 model. Thus, the suggested model is the finest technique among all models for Face Mask detection systems. Henceforth, the proposed model produced good results on the synthesized and the existing Face Mask datasets with a minimum loss, as shown in Table 4.

TABLE 5 Comparison of presented model and dataset with the existing approaches and datasets

Approaches	Accuracy (%) Existing Face Mask datasets: Face Mask dataset[29]	Medical Face Mask dataset[9]	Synthesized Face Mask dataset: FMDRT
Preeti Nagrath et al.[11]	92.48	93.02	95.6
Ruchi Jayaswal et al[28]	95.96	96.82	97.52
Xinqi et al.[1]	93.49	94.29	96.45
Arjya Das et al[34]	95.77	94.58	94.76
Md. Raffiuzzaman Bhuiyan[10]	93.45	94.26	96.32
Proposed model	97.86	96.93	98.75

5.5.1 | Comparative evaluation with the previous research methods

Consequently, we will compare the proposed results with the existing or past research methods in Table 5. As shown in Table 5, the suggested system is compared to various current techniques. Jiang et al.[1] have proposed a two-model Retina Face Mask that uses MobileNet and ResNet models to distinguish mask or no mask images. In addition, Preeti Nagrath et al.[11] used the MobileNet model with a single-shot detector for detecting masks, and no masks face images. Furthermore, the CNN pre-trained model computes another Face Mask detection approach by Arjya Das.[34] Similarly, Ruchi Jayaswal used the Inception model for Face Mask detection, and Raffiuzzaman recommended the YOLOv3 model to localize the Face Mask in a boundary box. These approaches are only limited to 2D Face Masks and surgical masks. Subsequently, the proposed model binds the pre-processing method with the CLAHE technique, encodes face detection using an SSD as a Face detector model, and classifies the Face Mask images by a deep transfer learning-based Xception network.

6 | CONCLUSION

This study has explored the Face Mask detection field. In this paper, a unique dataset is suggested, that is, (FMDRT) Face Mask Dataset in Real-Time, that is captured using webcams and mobile cameras. A novel technique named CL-SSDXcept is used to detect Face Masks in real-time. It generated outperforming results on synthesized and existing Face Mask datasets. The system was tested on the FMDRT dataset using five alternative models with distinct hyperparameter settings such as Learning_rate, batch size, and optimizers for training and evaluating the accuracy and computation time. These hyperparameters demonstrate how they influence the outcomes. On the FMDRT dataset, at 0.001 learning rate, 32 batch size, and ADAM optimizer produced encouraging results. The system predicted well ‘mask or no mask’ face images in real-time video frames and attained an impressive average 16 fps rate. This research found that the state-of-art model is lightweight in terms of model’s size with the best accuracy. Later, we also compared the results with existing Face Mask detection techniques and concluded that our proposed model produced fruitful results. In the future, this system can be helpful for more applications such as in hospitals, hazardous places where breathing is a significant issue, and many more.

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CONFLICT OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The datasets generated during and analyzed during the current study are available from the corresponding author on reasonable request. Please send requests for access to these data to ruchi.jayaswal@sitpune.edu.in. The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Ruchi Jayaswal https://orcid.org/0000-0002-1289-2160

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